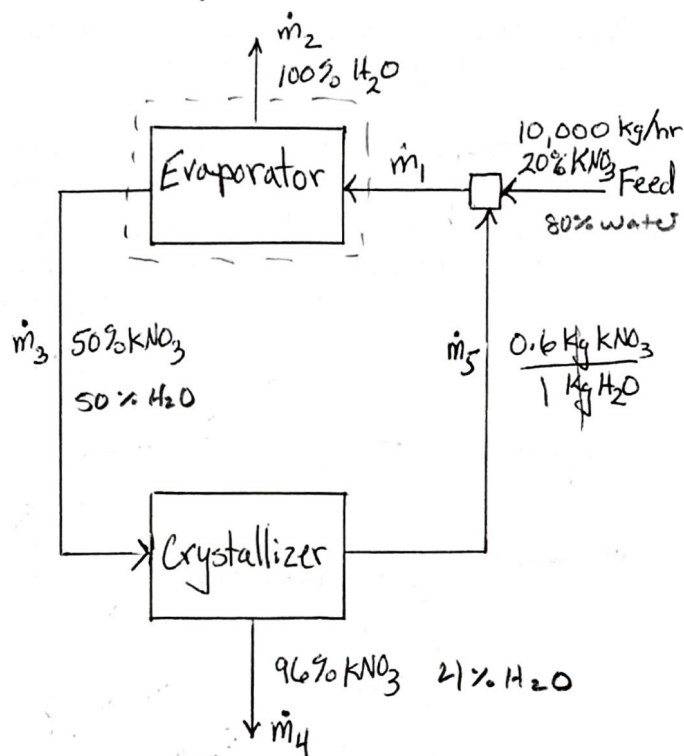


To receive full credit, you must show all of your work.

- (1) (25 pts.) Potassium nitrate (KNO_3) is separated as suspended solids from water (H_2O) by evaporation and then crystallization. A steady, continuous process is shown in the diagram. Calculate the five unknown mass flow rates $\dot{m}_1$, $\dot{m}_2$, $\dot{m}_3$, $\dot{m}_4$, and $\dot{m}_5$ in kg/hr. The feed enters at $\dot{m}_F = 10,000$ kg/hr and 20 wt% KNO_3 . Pure water leaves the top of the evaporator and a second stream of 50 wt% KNO_3 solution also leaves the evaporator. The crystallizer makes crystals that contain only 4 wt% water and a second saturated stream with 0.6 kg KNO_3 /kg H_2O which is recycled with the feed. Clearly indicate where you define your system boundaries for any balances you write. These systems should be carefully chosen to make the problem easier to solve.



Steady state, Find flows in kg/hr

$$.20 \cdot 10,000 \text{ kg/hr} = 2,000 \text{ kg/hr KNO}_3$$

$$.80 \cdot 10,000 \text{ kg/hr} = 8,000 \text{ kg/hr H}_2\text{O}$$

Balance around first box: $I_n = \text{Out}$

$$\cancel{\dot{m}_F} = \dot{m}_1 + \dot{m}_5$$

$$\text{KNO}_3 \text{ balance: } .20 \cdot \dot{m}_F = .6 \cdot \dot{m}_5 + \text{KNO}_3 \cdot \dot{m}_1 \rightarrow 2,000 \text{ kg/hr KNO}_3 = .6 \dot{m}_5 + \text{KNO}_3 \cdot \dot{m}_1$$

$$\text{H}_2\text{O balance: } .80 \cdot \dot{m}_F = .4 \dot{m}_5 + \text{H}_2\text{O} \cdot \dot{m}_1 \rightarrow 8,000 \text{ kg/hr H}_2\text{O} = .4 \dot{m}_5 + \text{H}_2\text{O} \cdot \dot{m}_1$$

Balance around Evaporator:

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \quad \checkmark$$

$$\text{KNO}_3 \text{ balance: } \text{KNO}_3 \cdot \dot{m}_1 = .5 \cdot \dot{m}_3 \quad \checkmark$$

$$\text{H}_2\text{O balance: } \text{H}_2\text{O} \cdot \dot{m}_1 = \dot{m}_2 + .5 \dot{m}_3 \quad \checkmark$$

Balance around Crystallizer:

$$\dot{m}_3 = \dot{m}_4 + \dot{m}_5 \quad \checkmark$$

$$\text{KNO}_3 \text{ balance: } .5 \dot{m}_3 = .6 \dot{m}_5 + .96 \dot{m}_4 \quad \text{No!}$$

$$\text{H}_2\text{O balance: } .5 \dot{m}_3 = .4 \dot{m}_4 + .04 \dot{m}_4$$

1	8
2	19
3	15
T	42

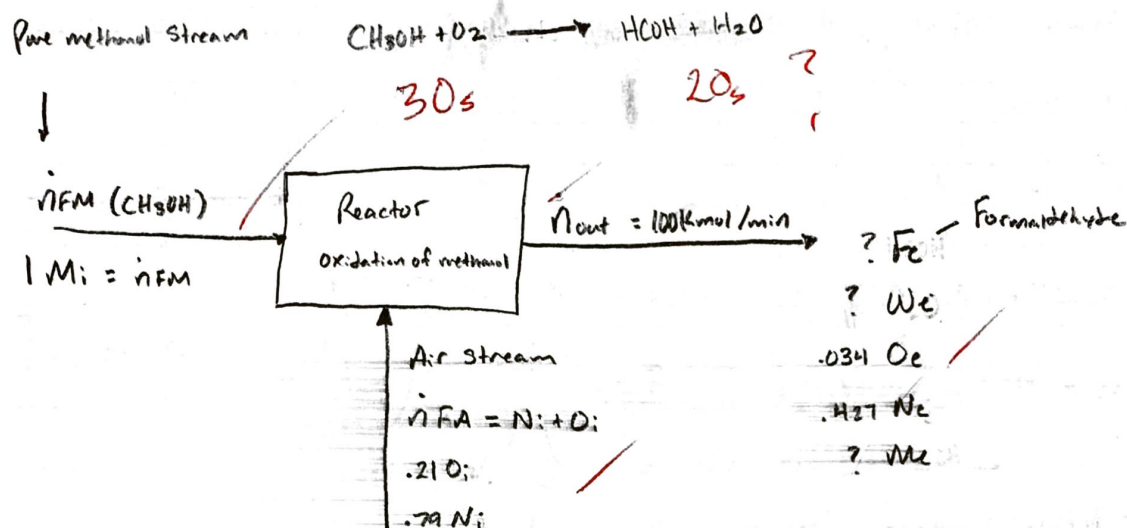
⑧

$\dot{m}_1 =$ _____ kg/hr, $\dot{m}_2 =$ _____ kg/hr, $\dot{m}_3 =$ _____ kg/hr,
 $\dot{m}_4 =$ _____ kg/hr, $\dot{m}_5 =$ _____ kg/hr

NAME William Isherwood

- (2) (45 pts.) Formaldehyde (HCOH) is produced by oxidation of methanol (CH₃OH). A reactor is fed a stream of pure methanol $\dot{n}_{FM}$ and a stream of air (21 mol% O₂ and 79 mol% N₂) $\dot{n}_{FA}$. A single product stream of 100 kmol/min leaves the reactor containing formaldehyde, water (H₂O), oxygen, nitrogen, and methanol. Chemical analysis determined that the product stream is 3.4 mol% oxygen and 42.7 mol% nitrogen.

- (a) Draw and label a block flow diagram of the process described above. Represent the species molar flow rates as N_i , O_i , and M_i for inlet nitrogen, oxygen, and methanol and N_e , O_e , M_e , W_e , and F_e for the exit nitrogen, oxygen, methanol, water, and formaldehyde. Notice that $\dot{n}_{FM} = M_i$ and $\dot{n}_{FA} = N_i + O_i$. Write a balanced chemical reaction equation for the process. What can you conclude about the mole fractions of the two reaction products [H₂O and HCOH] in the product stream? (10 pts.)



- H₂O and HCOH are produced in the reactor by the oxidation of methanol

- (b) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (5 pts.)

	Number	Explanation
4a. Stream Variables	8	1 component in $\dot{n}_{FM}$, 2 in $\dot{n}_{FA}$, and 5 in $\dot{n}_{out}$ = 8 total
4b. System Variable	1	There is one equation in the reactor
Total Independent Variables	9	Add up all the numbers from Part 4a and 4b
5a. Specified Flows	1	The amount of flows specified in the equation: 100 kmol/min
5b. Specified Stream Compositions	3	N-1 rule, only 1 can be used from the Air stream and 2 from $\dot{n}_{out}$
5c. Specified System Performances	0	There are no specified system performances
5d. Material Balance Equations	5	There can be up to 5 equations as there are 5 components in the system
Total Independent Equations	9	Add up all of the numbers from Part 5a to 5d
DOF	9-9=0	Subtract the total from 5a-5d from 4a-4b

(c) Using atomic balances, calculate the molar flow rates for the two feed streams in kmol/min for the methanol $\dot{n}_{FM}$ and the $\dot{n}_{FA}$. Also calculate the mole fractions of formaldehyde y_F , methanol y_M , and water y_W in the product stream. (30 pts.)

$$\dot{n}_{FM} + \dot{n}_{FA} = \dot{n}_{out}$$

$$\dot{n}_{out} = 100 \text{ kmol/min}$$

$$\text{Oxygen balance: } .21 \dot{n}_{FA_0} \times .034 \dot{n}_{out}$$

$$\dot{n}_{FA_0} = 16.19 \text{ kmol/min}$$

$$\dot{n}_{FA \text{ total}} = O + N:$$

$$\text{Nitrogen balance: } .79 \dot{n}_{FA_N} = .427 \dot{n}_{out}$$

$$\dot{n}_{FA_N} = 54.051 \text{ kmol/min}$$

$$\dot{n}_{FA \text{ total}} = 16.19 + 54.051$$

$$= 70.24 \text{ kmol/min}$$

No!

$$\text{Methanol balance: } \dot{n}_{FM} = M_e \cdot \dot{n}_{out}$$

Overall balance

$$\dot{n}_{FM} + \dot{n}_{FA} = \dot{n}_{out}$$

$$\dot{n}_{FM} + 70.24 \text{ kmol/min} = 100 \text{ kmol/min}$$

$$\dot{n}_{FM} = 29.76 \text{ kmol/min}$$

$$\text{Methanol balance: } \dot{n}_{FM} = M_e \cdot \dot{n}_{out}$$

$$29.76 \text{ kmol/min} = M_e \cdot 100 \text{ kmol/min}$$

$$M_e = 3.36 \text{ mol CH}_3\text{OH}$$

Water balance: Ran out of time to solve equation

but assume that Methanol to H₂O ratio is 1:4

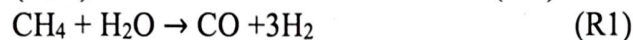
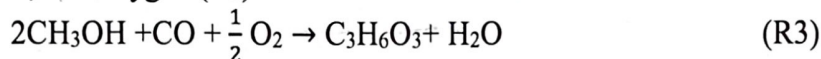
1 mol Methanol	4 mol H ₂ O
1 mol Methanol	

$$= 4 \text{ mol H}_2\text{O}$$

$$\sum y_i = 1$$

$$\dot{n}_{FM} = 29.76 \text{ kmol CH}_3\text{OH/min}, \dot{n}_{FA} = 70.24 \text{ kmol air/min}, y_F = \frac{7}{19} \text{ mol HCOC/mol},$$

$$y_M = 3.36 \text{ mol CH}_3\text{OH/mol}, y_W = \text{mol H}_2\text{O/mol}$$

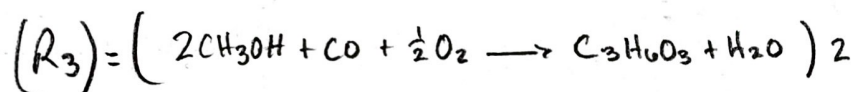
NAME William Isherwood(3) (30 pts.) A new synthesis mechanism for dimethyl carbonate (DMC, $C_3H_6O_3$) is as follows:Methane (CH_4) and water (H_2O) react to form carbon monoxide (CO) and hydrogen (H_2):CO and H_2 react to form methanol (CH_3OH): CH_3OH , CO, and oxygen (O_2) react to form DMC and water.(a) Use a generation-consumption analysis to synthesize a reaction pathway to DMC based on the three reaction above with no net generation or consumption of CO or CH_3OH . Hydrogen is allowed as a byproduct. Show your calculations and write the net chemical equation for the process. (20 pts.)

	R1	R2	R3	Net Rxn
CH_4	-3	0	0	-3
H_2O	-3	0	2	-1
H_2	9	-2	0	7
CO	+3	-1	-2	0
CH_3OH	0	1	-4	-3
O_2	0	0	-1	-1
$C_3H_6O_3$	0	0	2	2

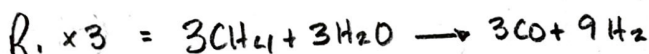
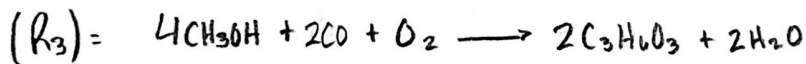
Positive is products and negative is reactants

net CO and net CH_3OH must be 0

zero!



12



NET RXN:

- (b) Calculate the fractional economy of the overall reaction. (Assume the atomic weights of carbon, hydrogen, and oxygen are: $M_C = 12$ g/mole, $M_H = 1$ g/mole, and $M_O = 16$ g/mole, respectively.) (10 pts.)

$$\text{fractional economy} = \frac{\text{total weight of elements}}{\text{molar mass of element}}$$

3

not correct but assume there are 6 Carbons in net reaction, 410 Hydrogens, and 9 oxygens

Fractional atom economy = _____

15